

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester II Examination, 2024-25 (NEP)

PHYSICS (SEC)**Paper Code: PHYSE21 : Energy Conversion and Storage for Sustainable Development**

Ref. No.: CECNF/N/2024-25/6

Time: 2:30 Hours

Maximum Marks: 70

*Instructions: Attempt all questions (internal choices are provided).***Question 1.** Attempt *any four* of the following: [4 × 3.5 = 14]

- (a) Write four differences between a battery and a fuel cell?
- (b) Give the definitions of volumetric and gravimetric density. In a typical high pressure hydrogen storage cylinder, how do gravimetric hydrogen density change with increasing pressure?
- (c) What are the objectives of National Green Hydrogen Mission?
- (d) What is the meaning of net-zero emission?
- (e) What is quantum efficiency? Describe the spectral response of a photovoltaics. Describe the mechanism of low efficiency at red and blue side.
- (f) Name the major types of energy storage technologies and give example of each technology.
- (g) Write down the challenges in developing a thermal energy storage plant.
- (h) What is a flow battery? What are the drawbacks of Nickel-based batteries?

Question 2. Describe the hydrogen liquefaction process using Joule-Thompson cycle. What could be some possible applications for liquid hydrogen storage systems and why? What is physisorption of hydrogen? [7+3+4]

OR

Give examples of PV materials. What is P-N junction diode? What are forward and reverse bias characteristics of a P-N Junction diode? Describe different parameters in an I-V characteristics of a photovoltaics. [3+3+4+4]

Question 3. Under National Solar Mission, India has launched PM Surya Ghar Yojana (PMSGY) to incentivise the use of solar energy. Describe the benefits available under this scheme. What are the components that you need to build an off-grid solar system for your home? The cost of which ones of the components are covered by PMSGY? What are the differences between on-grid and off-grid solar systems? [4+3+3+4]

OR

What do you understand by primary and secondary batteries? Explain with example. Give three examples of commercially used secondary batteries and compare their functioning. What is the Solid Electrolyte Interface (SEI)? Explain its importance in energy devices. [4+6+4]

Question 4. Describe the working principles of Dye-Sensitized Solar Cells (DSSC) which can harvest energy from indoor lightings. List some of the current challenges to develop high efficiency and low cost solar cells. [7+7]

OR

Briefly describe three underground compressed air storages. What are salt domes? Describe how lake and oceans could be used as compressed air storage. What are the thermodynamic processes typically used in compressed air storage technologies? [4+2+4+4]

Question 5. What is a Wind Energy Conversion System (WECS)? Describe the mechanical and electrical components of this system. What are horizontal and vertical axes of WECS? What are active and reactive power control systems in a WECS? Describe the challenges one would face in planting an WECS. [2+5+2+2+3]

OR

What do you understand by the sustainable development in terms of energy devices? Discuss the future possible energy storage technologies which are being researched to attain a better sustainable development. [7+7]
